
Imitating Human Hand Manipulation of Objects with PPO in Isaac Lab

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Abstract

We train a floating base Shadow Hand to imitate how a human hand manipulates objects, using motions from the HO-Cap dataset and PPO in Isaac Lab. The given reference comes from video reconstruction, so it is not physically real: the hand can pass through the object, or float in the air without really holding it. Our task is to find a physically valid motion that stays close to this reference. We design an observation that tells the policy the tracking error to both the current and the next reference frame, and a reward made of several bounded terms. The two main ideas in the reward are (i) grasp targets that are attached to the object, so the human grip is kept fixed to the simulated object as it moves, and (ii) a thumb opposition term that only pays when the thumb and an opposing finger press the object at the same time. We use one shared design for three sequences (one main and two optional), including a thin flat object that has to be picked up, turned about 180 degrees, and put down. The policy follows the reference hand within about 1.8 to 2.7 cm and the object within about 4.5 to 8.9 mm. Two small config changes (stronger smoothing and tighter pose tracking) also remove finger shaking and an over curled index finger, and they improve the numbers.

1 Introduction

The goal is to train a reinforcement learning agent that moves an object by following the hand motion in a human video. We use the HO-Cap dataset. For each frame it gives the human hand keypoints, the object pose, and the object velocity, all built from multi-view capture and kinematic reconstruction. Because the data is kinematic, it is not always physically valid: the reconstructed hand can pass through the object or float without really grasping it. Our objective is to use Isaac Lab, a physics simulator, to get a physically valid motion of the hand and the object that still stays close to the demonstration.

The PPO algorithm (rl_games) is given and we do not change it. The given environment provides the action interface and the data loading, but it leaves the observation, the reward, and the intermediate values empty. These are the parts we design. The result is judged by how closely the robot hand motion and the object motion (position and rotation) match the reference. We report three sequences: a **main** sequence (Sequence 1) and two **optional**, harder sequences (Sequences 2 and 3), shown in Figure 1. One shared design is used for all three, and the sequence is chosen at run time by an environment variable.

Given data. Per frame: 21 human hand keypoint positions ($T \times 21 \times 3$), object position ($T \times 3$), object rotation as a wxyz quaternion ($T \times 4$), and object linear and angular velocity ($T \times 3$). The robot is a Shadow Hand. Its floating base (the wrist) is driven by an external force and torque, and its fingers are position controlled. The action has 9 base values (translation plus 6D rotation) and the actuated finger joints.

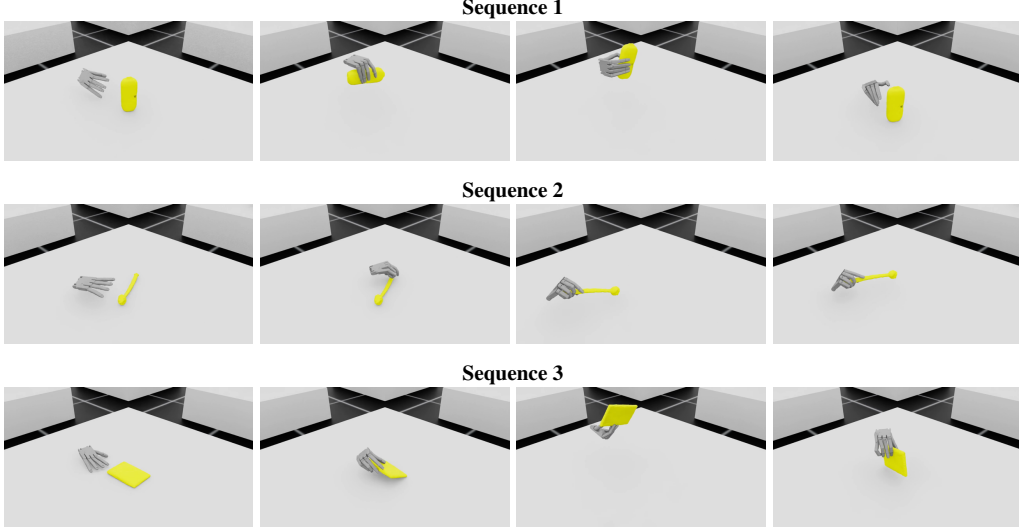


Figure 1: The three sequences (frames from the learned rollout, time runs left to right). Seq. 1 is a tall upright object, Seq. 2 a long thin rod, and Seq. 3 a thin flat piece lying on the table. Full videos are in seq1.mp4, seq2.mp4, and seq3.mp4.

Table 1: What the policy sees at each step (284 values). *Reference tracking* is the error to the demonstration, given for the current and the next frame; it is the largest group and the most useful one for imitation. The errors are vectors, so the policy also sees which way to move.

Group	What it contains (dimension of each part)	Dim
Hand state	joint angles (22), joint speeds (22), wrist pose and speed (15)	59
Object state	position (3), rotation (6), speed (6), hand-to-object (3)	18
Reference tracking	hand keypoint error, cur + next (63+63); object pose error, cur + next (18)	144
Grasp features	fingertip-to-target (15), fingertip-to-object (15)	30
Contact and memory	contact forces (5), previous action (27), phase (1)	33
Total		284

2 Method

We use one shared environment for all sequences. Only the reference data path changes between sequences, which also changes the object and the trajectory. Below we describe the observation, the reward, the helper components, and finally two small changes that improve the motion quality. The full implementation is included in the submission package.

2.1 Observation

The observation is a vector of 284 values (Table 1). The main idea is to give the policy the tracking error to both the current and the next reference frame. This way the policy always knows where the reference hand and object are now, and where they will be next, which makes it much easier to follow a moving target.

2.2 Reward

Every tracking term is a bounded exponential of the form $w \exp(-k e) \in (0, w]$. This way no single term can take over the total, and the gradient is good near zero error. Let p_i and $\hat{p}_i$ be the robot and reference hand keypoints, $o, \hat{o}$ the object positions, and $q, \hat{q}$ the object orientations. The tracking

Table 2: Reward terms. Each tracking term has the bounded form $w \exp(-k e)$ of its error e ; a larger rate k means a stronger push toward low error. We also use small penalties on the action size (-0.004) and the joint velocity (-0.001), a smoothing factor $\alpha = 0.3$, and a contact scale $f_0 = 0.5$ N.

Term	what it measures	weight w	rate k
r_{kpt}	hand keypoint error	1.0	30
r_{pos}	object position error	1.0	20
r_{rot}	object rotation error	1.0	3
r_{grasp}	fingertip to target error	0.8	20
r_{con}	thumb opposition contact	0.5	n/a

terms are

$$r_{\text{kpt}} = w_{\text{kpt}} \exp\left(-k_{\text{kpt}} \frac{1}{21} \sum_i \|p_i - \hat{p}_i\|\right), \quad r_{\text{pos}} = w_{\text{pos}} \exp\left(-k_{\text{pos}} \|o - \hat{o}\|\right), \quad (1)$$

$$r_{\text{rot}} = w_{\text{rot}} \exp\left(-k_{\text{rot}} d_{\text{geo}}(q, \hat{q})\right), \quad r_{\text{grasp}} = w_{\text{grasp}} \exp\left(-k_{\text{grasp}} e_{\text{grasp}}\right), \quad (2)$$

where d_{geo} is the geodesic angle between the two orientations and e_{grasp} is the mean fingertip distance to its target attached to the object (defined next).

Grasp targets attached to the object. For each frame the reference gives the human fingertip positions relative to the object center, δ_i . Instead of tracking these in the world frame, we fix them to the simulated object. The target for fingertip i is

$$\tau_i = o + R(q \hat{q}^{-1}) \delta_i, \quad e_{\text{grasp}} = \frac{1}{5} \sum_i \|\tau_i - p_i^{\text{tip}}\|, \quad (3)$$

so the human grip follows the object even when the object has moved or turned away from the reference pose. This keeps the fingers in the same grip relative to the real object, instead of chasing a fixed point in the world.

Thumb opposition contact. A simple contact reward (any finger touching) usually makes the hand press from one side, which cannot control the object. So we reward opposition instead. Let $c_{\text{th}}, c_{\text{op}} \in [0, 1]$ be the thumb contact and the best opposing finger contact, both scaled by force. The contact term is turned on by a per frame flag $g_t \in \{0, 1\}$ that is 1 only when the reference hand is holding the object (Section 2.3):

$$r_{\text{con}} = w_{\text{con}} g_t \left[0.15 (c_{\text{th}} + c_{\text{op}}) + 0.7 \min(c_{\text{th}}, c_{\text{op}}) \right]. \quad (4)$$

The min term pays only when the thumb and an opposing finger press at the same time, which leads to a stable grip the hand can control. The small added term gives a little reward from the first touch, so the policy can start to learn.

Total reward. With small penalties on the action p_{act} and on the joint velocity p_{dof} , the reward per step is

$$r = \max(0, r_{\text{kpt}} + r_{\text{pos}} + r_{\text{rot}} + r_{\text{grasp}} + r_{\text{con}} + p_{\text{act}} + p_{\text{dof}}). \quad (5)$$

We clamp it to be non-negative, so ending the episode early is never better than continuing. Table 2 lists the weights.

2.3 Helper components

Contact gate. The gate g_t is computed once from the reference. A frame is a grasp frame when the thumb and an opposing fingertip are both within 10 cm of the object center. This keeps the contact reward only in the part where the human is really holding the object.

Early termination. During training only, the episode ends if the object goes too far from the reference ($\|o - \hat{o}\| > 10$ cm) or the hand flies away (mean keypoint error > 15 cm), after a short grace of 10 steps. Together with the non-negative reward, this makes it better for the policy to stay close to the reference.

Control smoothing. The finger targets are smoothed by an exponential moving average with weight α on the new action, and the wrist force and torque are smoothed in the same way. This is a smoothing of the control, not a reward term.

2.4 Two changes for motion quality

The base design already grasps and follows the reference, but the played motion has two small problems that are clearest in the videos: the fingers shake at high frequency, and the index finger curls in too much during the grasp. We do not fix these with action penalties. In our tests such penalties made the early exploration too careful and stopped the opposition grasp from forming at all. Instead we use two settings that cannot block the grasp from being found. First, we smooth the fingers more ($\alpha : 0.5 \rightarrow 0.3$), which is only a smoothing of the control. Second, we make the pose tracking sharper ($k_{\text{kpt}} : 20 \rightarrow 30$), which pulls all 21 keypoints more strongly toward the reference. Since the keypoint error is driven by the worst finger, this push lands mostly on the index finger that curls. Both changes improved the numbers instead of hurting them.

3 Results

Setup. We train with PPO (rl_games), 1024 parallel environments, 5000 epochs, and a fixed seed, one run per sequence with the same shared code.¹ The curves below show the raw signal (faint) and a moving average (bold). For each sequence we show the episode return, the episode length, the contact opposition, and the three tracking errors (hand keypoint, object position, object rotation). The final numbers are in Table 3.

3.1 Sequence 1 (main): an upright object, grasp and reorient

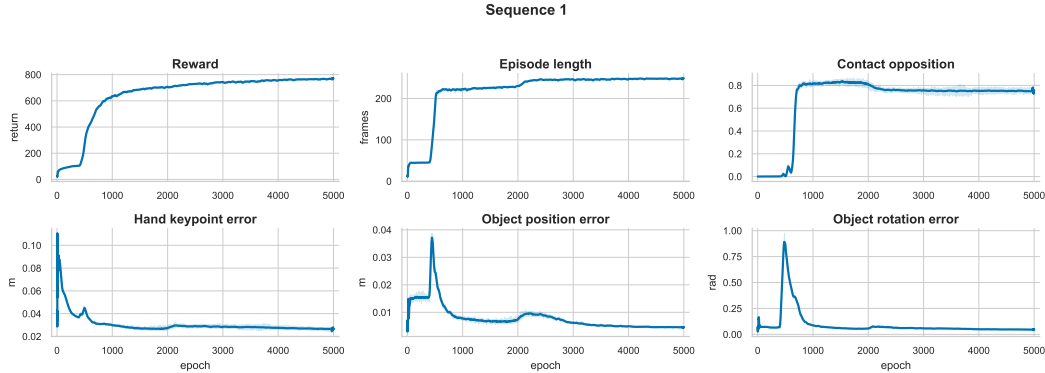


Figure 2: Sequence 1. The early rise of the object rotation error and its sharp drop mark the moment the hand stops pushing the object and starts to hold it (around epoch 600, when opposition rises). After that all errors stay low and the hand finishes the full episode.

Sequence 1 is a tall, upright object (about $20 \times 8 \times 5$ cm, long along its vertical axis) that the hand grasps and then turns. Its curves are the cleanest of the three. The policy first explores, then grasps, then tracks. For about the first 500 epochs the policy only pushes the object: the return stays near 100, the opposition is near zero, and the object rotation error climbs to about 0.9 rad. The reason is that the reference is already turning the object while the hand cannot hold it yet, so the simulated object falls behind the turning reference.

The key moment happens around epoch 500 to 800, and several signals change at once. The opposition jumps from about 0 to about 0.7 (the thumb and an opposing finger touch at the same time), the object rotation error drops from about 0.9 to about 0.05 rad, and the return jumps from about 100 to about 600. This is the moment the opposition grasp is found and the hand starts to control the object. After that every error stays low and flat, and the episode runs almost to the full 250 frames

¹The submission is a single package: one shared implementation (`gr_env.py` and `gr_env_cfg.py`, identical for every sequence) plus, under each `seqN/` folder, the best checkpoint (`shadow_hand_train.pth`) and a playback video (`seqN.mp4`). Any result reported here can be reproduced with `GR_SEQ=N CKPT=seqN/shadow_hand_train.pth bash run_play.sh`.

(248 of 250). The final errors are good (2.7 cm hand, 4.5 mm position, 0.047 rad which is about 2.7 degrees). This rise and sharp drop is the clearest sign in all our runs of the change from pushing to holding that the opposition term is meant to cause (seq1.mp4).

3.2 Sequence 2 (optional): a long thin object, grip and roll

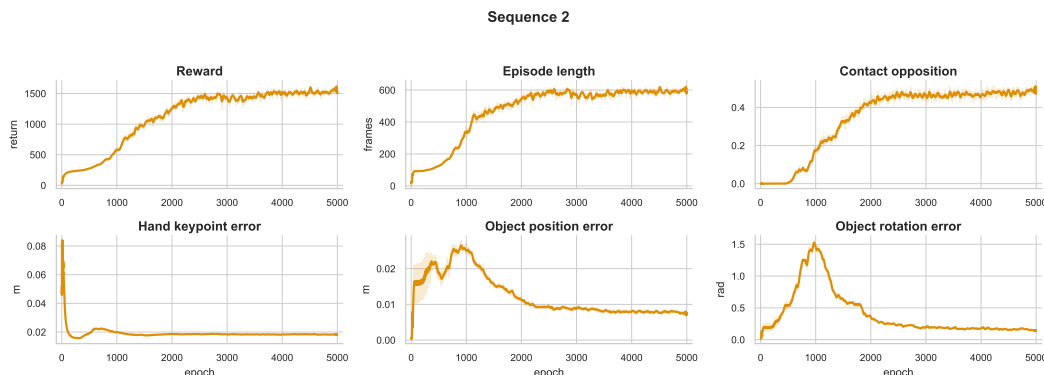


Figure 3: Sequence 2. The grip moves down the shaft as the rod shaped object is rolled, so the opposition stays low (about 0.49) even though the object position and rotation are followed well.

Sequence 2 is a long, thin rod shaped object with a blunt end (about $28 \times 5 \times 4$ cm, so it is roughly 7 times longer than it is thick). The hand grasps it near the tip and then grips lower and lower along the shaft while the object is rolled. Unlike Sequence 1, the curves show no sharp grasp moment. The return, the episode length, and the opposition all rise slowly over several thousand epochs, because there is no single point where the object becomes easy to control. The grip has to be made again and again as the hand slides down the shaft and the object rolls.

This moving grip is also why the opposition settles at only about 0.49, the lowest of the three. A fixed two finger pinch is never held for long, so the opposition at each moment stays in the middle, even though the object itself is followed well. The object position error has a large early bump. It reaches about 2.5 cm near epoch 1000, when the thin object slips easily, and then it drops to 7.3 mm. The object rotation error also rises to about 1.5 rad before it falls to 0.14 rad. The part that is left shows how hard a constant roll is to match exactly. In the end the policy survives 602/660 frames. So Sequence 2 is mostly about a grip that keeps changing, and the smooth curves are the sign of that slow skill (seq2.mp4).

3.3 Sequence 3 (optional): a thin flat object, lift and flip

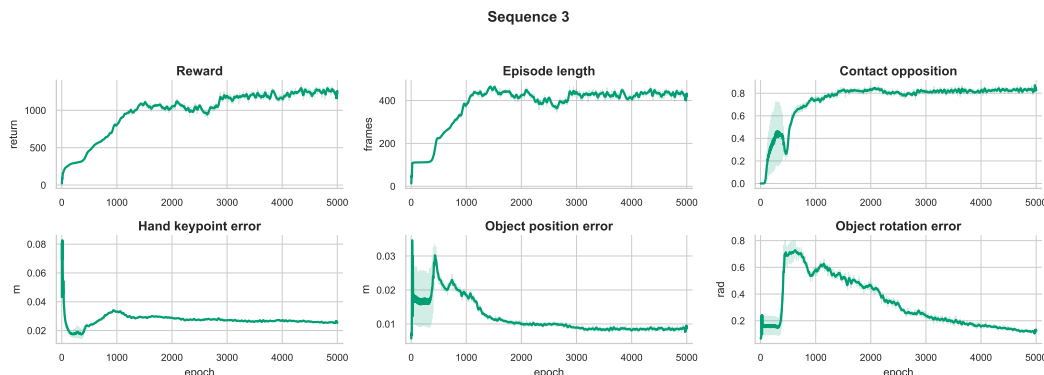


Figure 4: Sequence 3. The episode length rises from about 110 to about 415 frames, and at the same time the object rotation error falls from about 0.7 to 0.13 rad. Together they show the policy learning to lift and flip the flat object.

Table 3: Final results (mean over the last 20 logged points), each measured from the submitted per-sequence best checkpoint. The three error columns are the hand and object tracking errors; lower is better. `opp.` is the thumb opposition contact (0 to 1, higher is firmer) and `len.` is the episode length out of the full sequence length.

Sequence	return	kpt err (m)	obj-pos (m)	obj-rot (rad)	opp.	len.
1 (main)	770	0.0268	0.0045	0.0468	0.74	248/250
2 (optional)	1565	0.0183	0.0073	0.1405	0.49	602/660
3 (optional)	1205	0.0259	0.0089	0.1253	0.84	415/510

Sequence 3 is the hardest. The object is a thin, wide flat piece (about $19 \times 14 \times 2.6$ cm) lying flat on the table, so its center starts only about 1.5 cm above the surface. The reference motion is a lift and flip. The object is picked up, turned about 180 degrees while it is carried about 24 cm, and then put back down. Getting the fingers around a flat object on the table and then turning it a full half turn is much harder than the other two sequences, and the learning happens in two clear stages.

In the first stage (up to about epoch 400) the episode length is stuck at about 110 frames and the return stays near 300. The policy cannot get under the flat object to lift it, so as soon as the reference starts to rise (around frame 88) the object falls behind, its position error passes the termination limit, and the episode is cut short. The opposition even drops to about 0.27 during this struggle. In the second stage the policy gets through. The episode length rises steadily from about 110 to about 415 frames, and the return rises from about 300 to about 1205, as the policy learns to lift and carry the object further along the trajectory. The object rotation error shows the same thing for the flip. It rises to about 0.7 rad as the half turn starts, then it is slowly reduced to 0.13 rad as the turn is learned. At the same time the opposition comes back to 0.84, the strongest grip of the three, which is the firm hold a flat object needs to be carried and flipped. The episode length curve is the most useful signal here, because it directly shows the lift being learned. The fact that the same shared design learns this much harder task with no special change shows that the reward (object attached targets plus opposition) works for very different objects (seq3.mp4).

4 Discussion and Conclusion

One shared design (bounded tracking terms, grasp targets attached to the object, and a thumb opposition reward that is on only during the reference grasp) works across three very different objects. It grasps and turns an upright object, rolls a long thin object, and lifts and flips a thin flat object, while keeping the hand error within a few centimeters and the object error a few millimeters. The two most useful choices were the targets attached to the object, which keep the grip matched to the moving object, and the opposition term, which gives a grip the hand can control instead of a one sided push. For motion quality, we found that the small problems should be fixed without action penalties that block exploration. More smoothing and tighter pose tracking reduced the shaking and the over curled index finger, and they also improved the numbers. Still, some finger trembling is left. The stronger control smoothing made it smaller but did not remove it fully, so there is room to improve here. Because a smoothing penalty added too early blocks the opposition grasp from forming, one safe option is to phase in a small penalty on the action change only after the grasp is stable, so it cleans up the motion without hurting grasp discovery. The other main limit is the object rotation error on the most dynamic sequences, mostly from the 180 degree flip in Sequence 3. A curriculum that relaxes the termination during large turns would likely help.